

Soft-Margin SVM: Slack, Hinge Loss and C

Contents

1. Overview	2
2. Why the hard margin is not enough	2
2.1 One outlier, in one dimension	3
2.2 The same failure in the formula	4
3. From maximising to minimising	4
4. Slack: how far a point breaks the rules	6
4.1 The new term	6
4.2 What ξ measures	6
5. The soft-margin loss	7
6. The hyperparameter C	9
7. Comparison with logistic regression	10
8. Why “soft margin”	11
9. Summary	12
10. Sources	12
11. Key terms	12

Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Classification problems ([Note ML-003](#)); Regularisation ([Note ML-062](#)); Perceptron trick ([Note ML-069](#)); Kernel trick ([Note ML-089](#)); Dot product ([Note MA-050](#)); Equation of a hyperplane ([Note MA-051](#)).
- **Compare with:** Log loss (binary cross entropy) ([Note ML-072](#)); Logistic regression ([Note ML-074](#)); Perceptron loss ([Note DL-006](#)).

1. Overview

Key point: Soft-margin SVM lets some points break the rules, but charges for it. Its loss adds a margin error and C times a classification error; C sets the balance.

The hard-margin SVM of the [SVM maths Note](#) needs data that a hyperplane separates perfectly. Real data often is not (ISL §9.1.5). The **soft-margin SVM** (G-1829) fixes this by giving every point a little room to be wrong, and adding the cost of that room to the loss (Figure 1).

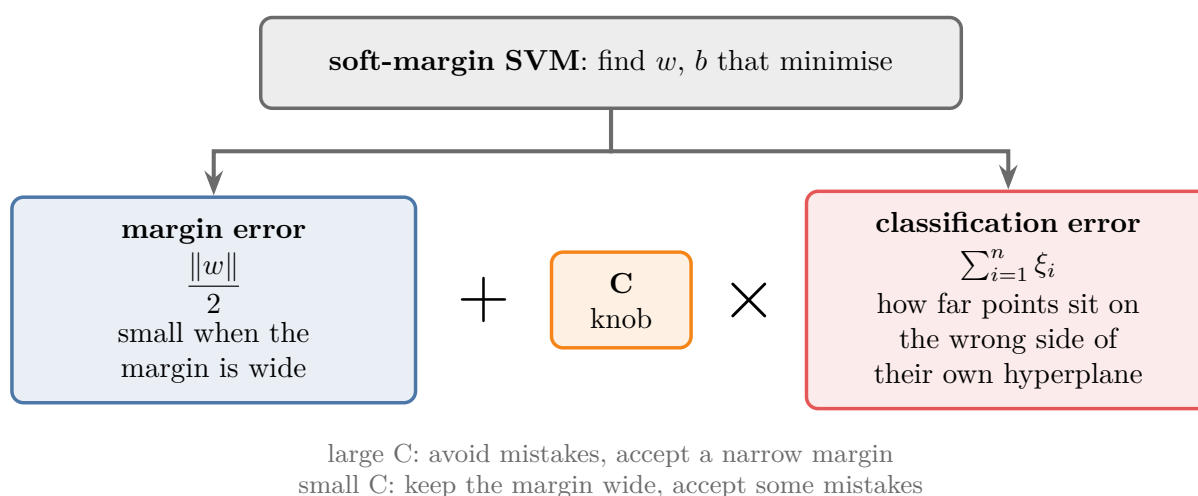


Figure 1: The soft-margin loss: a margin error plus C times a classification error

2. Why the hard margin is not enough

Key point: The hard-margin constraints allow no exceptions, so a single point on the wrong side leaves the problem without a solution.

2.1 One outlier, in one dimension

Key point: One outlier drags the hard-margin threshold right next to the other class. Allowing that one mistake puts the threshold back between the two main groups.

Take the iris petal lengths of the [SVM intuition Note](#): setosa petals up to 1.9 cm, versicolor petals from 3.0 cm, and the best threshold in the middle of the gap. Now add one made-up outlier: a flower labelled setosa with a 2.9 cm petal, right next to the versicolor flowers (the diamond in Figure 2).

1. **No mistakes allowed.** The threshold must keep the outlier on the setosa side, so it is squeezed between 2.9 and 3.0 cm: threshold 2.95 cm, margin 0.1 cm. A new flower with a 2.8 cm petal is then called setosa, although it sits next to the versicolor flowers and 0.9 cm from the nearest ordinary setosa.
2. **One mistake allowed.** Give up the outlier and the threshold moves back to 2.60 cm, between the two main groups, with a margin of 1.4 cm. The outlier is now misclassified, but the new flower is called versicolor, which is what its neighbours say.

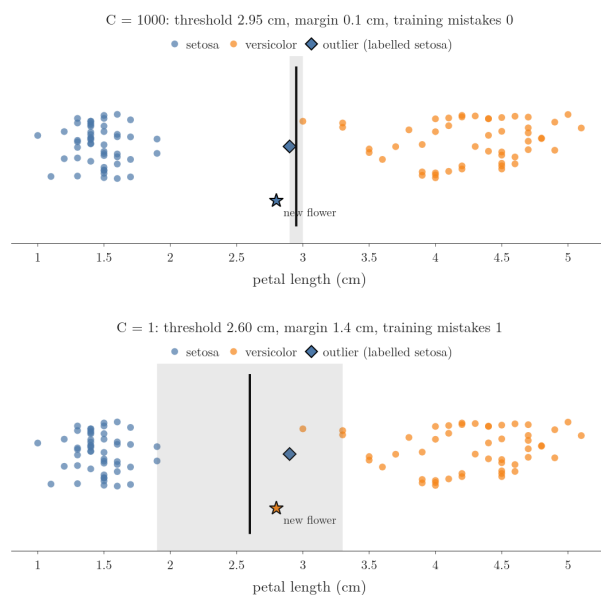


Figure 2: Iris petal lengths with one made-up outlier (diamond), and a linear SVM as the price of a mistake, C , falls from 1000 to 1. The threshold (black) moves from 2.95 cm, squeezed against the versicolor flowers, back to 2.60 cm; the margin (grey band) grows from 0.1 to 1.4 cm. The new flower (star) takes the colour of its predicted species. Idea after StatQuest, “Support Vector Machines Part 1 (of 3): Main Ideas!!!”, shown there on mouse weights.

In Figure 2, watch the black threshold and the grey margin as mistakes become cheaper. The number C in the titles is the price of a mistake; Section 6 defines it.

The second threshold is wrong on one training flower but should do better on new flowers. In the language of the [bias-variance trade-off](#) (G-288): allowing a mistake adds a little bias, and

removes a lot of variance, because one unusual point no longer decides where the threshold goes. A margin that is allowed to contain or misclassify points is a **soft margin**. How many mistakes to allow is chosen by cross-validation (Section 6).

2.2 The same failure in the formula

Key point: With one constraint per point and no exceptions, a single point on the wrong side leaves the problem without a solution.

The hard-margin SVM solves the problem below. Its symbols, each with a value from the best line of the [SVM intuition Note](#):

- $w = (0.049, 0.898)$ is the weight vector (one number per feature) and $b = -4.485$ the intercept;
- $\|w\|$ is the length of w : $\sqrt{0.049^2 + 0.898^2} = 0.899$;
- $w^T x_i$ multiplies the weights by the features of point i and adds: for $x_i = (2.5, 3)$ it is $0.049 \times 2.5 + 0.898 \times 3 = 2.82$;
- y_i is the class of point i , $+1$ or -1 , and n is the number of points (16 here).

The hard-margin problem:

$$\arg \max_{w, b} \frac{2}{\|w\|} \quad \text{such that} \quad y_i (w^T x_i + b) \geq 1 \quad \text{for all } i$$

Each point is one **observation** (G-1374; one record, a row of the data table). Its coordinates are its **features** (G-772; input variables, one column each), and its class y_i ($+1$ green, -1 red) is the **target** (G-1949; the output we predict).

The constraint says that every green point lies on or above π^+ and every red point on or below π^- . A green point below π^+ breaks the constraint: its target is $+1$ but $w^T x_i + b$ is less than 1, so the product is less than 1. A red point above π^- breaks it the same way, with $y_i = -1$ and $w^T x_i + b > -1$.

The hard-margin problem is a constrained optimisation problem with n constraints, one per point. The problem assumes that **no** point ever lies inside the margin or on the wrong side. Many real datasets are at best almost linearly separable: a few points, often outliers, sit on the wrong side. With such data the constraints cannot all be met (Figure 7 of the SVM maths Note), and the hard-margin problem has no solution (ISL §9.1.5).

We need a version that leaves some space for outliers. That version is the soft-margin SVM.

3. From maximising to minimising

Key point: Maximising $2/\|w\|$ is the same as minimising $\|w\|/2$. The minimising form is easier to extend.

Before changing anything, we rewrite the hard-margin problem. Take three candidate lines whose margin f is 1, 2 and 4. Their inverses $1/f$ are 1, 0.5 and 0.25. The line with the

largest f (4) has the smallest $1/f$ (0.25). In general, for a positive quantity f , the largest f is exactly where $1/f$ is smallest, so maximising f and minimising $1/f$ give the same answer. Here $f = 2/\|w\|$, so $1/f = \|w\|/2$:

$$\arg \max_{w,b} \frac{2}{\|w\|} = \arg \min_{w,b} \frac{\|w\|}{2}$$

With numbers, for the best line of the SVM intuition Note, $\|w\| = 0.899$:

$$\text{margin} = 2/0.899 = 2.22$$

$$\|w\|/2 = 0.899/2 = 0.45$$

The margin 2.22 is as large as possible exactly when 0.45 is as small as possible. The rewritten problem is still the hard-margin SVM; only the form has changed, because a minimisation is easier to add terms to.

Figure 3 shows why the two forms agree. As $\|w\|$ grows, the margin $2/\|w\|$ falls and the term $\|w\|/2$ rises, so the smallest $\|w\|$ the constraints allow is the best choice for both.

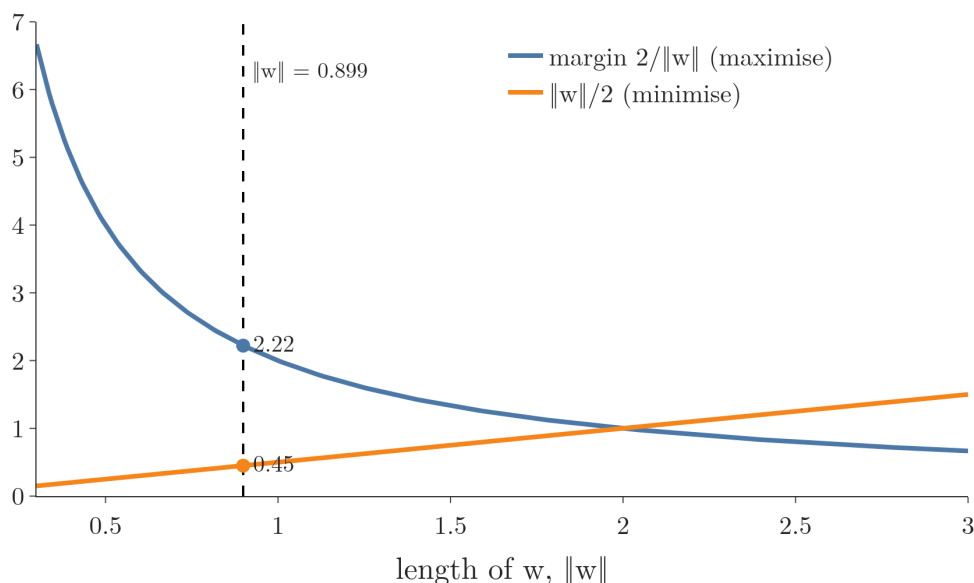


Figure 3: The margin $2/\|w\|$ (blue) and the term $\|w\|/2$ (orange) against the length of w . The first falls where the second rises, so the same $\|w\|$ makes the margin largest and the term smallest; marked: $\|w\| = 0.899$, margin 2.22, term 0.45.

Extra: In practice, and in scikit-learn, the term is written $\frac{1}{2}\|w\|^2$ instead of $\|w\|/2$ (scikit-learn user guide §1.4.7). Squaring does not change which w is smallest, and the squared form is smooth (it has no kink at $w = 0$) and is exactly the L2 penalty of [Ridge](#)

regression.

4. Slack: how far a point breaks the rules

Key point: Each point gets a slack ξ_i . The slack is 0 for a point on the correct side of its own hyperplane, and grows with how far the point sits on the wrong side of it.

4.1 The new term

Key point: Add $\sum \xi_i$ to the loss: the total amount by which the points break their constraints.

To make the SVM soft, we add one more term to the minimisation:

$$\arg \min_{w,b} \frac{\|w\|}{2} + \sum_{i=1}^n \xi_i$$

The Greek letter ξ is “xi”. Each training point i gets its own value ξ_i , called its **slack** (G-1820). The symbol $\sum_{i=1}^n$ (sigma) means “add up the terms for $i = 1, 2, \dots, n$ ”. On a toy set of $n = 4$ points with slacks $\xi_1 = 0$, $\xi_2 = 1.33$, $\xi_3 = 0$, $\xi_4 = 1.32$:

$$\sum_{i=1}^4 \xi_i = 0 + 1.33 + 0 + 1.32 = 2.65$$

The symbol $\arg \min$ is the mirror of $\arg \max$: it returns the w and b that give the smallest value of the expression. The algorithm now looks for the w and b that make **both** terms small.

4.2 What ξ measures

Key point: During training, green points are judged against π^+ and red points against π^- , not against π .

During training, each class is judged against its **own** hyperplane: green points against π^+ , red points against π^- . (The middle line π is only used later, to classify new points.)

- A green point on or above π^+ , or a red point on or below π^- , is fine: $\xi_i = 0$.
- A green point below π^+ , or a red point above π^- , is an error point. Its ξ_i says how far it is from its own hyperplane, in the wrong direction. The unit is the margin itself: the gap from π to π^+ counts as 1. So $\xi_i = 1$ means the point sits exactly on π , and $\xi_i > 1$ means it has crossed to the wrong side of π .

So an error point may be inside the margin but still on the correct side of π , or right across π on the wrong side. The further it is from where it should be, the larger its ξ_i .

Figure 4 shows the two error points of a soft-margin SVM. The green point (6, 4.3) lies below π^+ , even below π : $\xi = 1.33$. The red point (2.8, 5.2) lies above π^- and above π : $\xi = 1.32$. Every other point has $\xi = 0$.

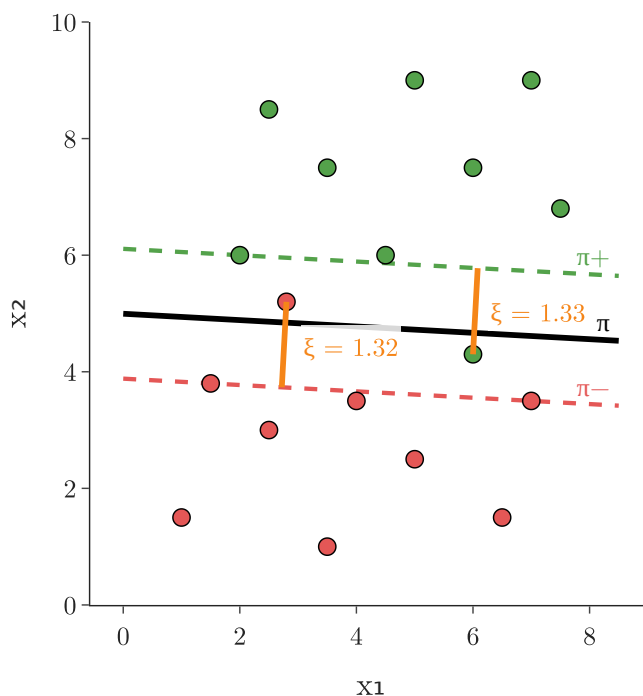


Figure 4: The slack ξ of the two points that break their constraints (orange); all other points have $\xi = 0$

Minimising $\sum \xi_i$ therefore means keeping the errors few and small.

Extra: Precisely, the slack is $\xi_i = \max(0, 1 - y_i(w^T x_i + b))$, and the hard constraint is loosened to $y_i(w^T x_i + b) \geq 1 - \xi_i$ with $\xi_i \geq 0$. The distance in the plot, in the units of the axes, is $\xi_i / \|w\|$: for the green point, $1.33 / 0.900 = 1.48$. A point with $0 < \xi_i \leq 1$ is inside the margin but still correctly classified; $\xi_i > 1$ means it is on the wrong side of π .

5. The soft-margin loss

Key point: Loss = margin error + $C \times$ classification error. The first term wants a wide margin, the second wants few and small mistakes.

The two terms pull in different directions:

- $\|w\|/2$ is the **margin error** (G-1161). The margin error equals $1/d$, so it falls as the margin widens: a wide margin gives a small margin error.

- $\sum \xi_i$ is the **classification error** (G-393). The classification error falls as the error points move back to their own side.

The last step is a weight on the classification error, a positive number C . In words: find the line that balances a wide margin against small errors, with C deciding how much the errors count. As a formula, the soft-margin SVM solves:

$$\arg \min_{w,b} \underbrace{\frac{\|w\|}{2}}_{\text{margin error}} + C \underbrace{\sum_{i=1}^n \xi_i}_{\text{classification error}}$$

With numbers, for the line of Figure 4. First the two parts, which do not depend on C :

$$\text{margin error} = \|w\|/2 = 0.900/2 = 0.45$$

$$\text{classification error} = \sum \xi_i = 1.33 + 1.32 = 2.65$$

With $C = 1$ the loss is:

$$0.45 + 1 \times 2.65 = 3.10$$

With $C = 10$ the same mistakes cost more (the unrounded $\sum \xi_i$ is 2.652):

$$0.45 + 10 \times 2.652 = 0.45 + 26.52 = 26.97$$

A larger C makes each mistake cost more, which is why a larger C pushes the line to fix them. Figure 5 stacks the two parts for both values of C : the margin error is a thin slice either way, and the classification error takes over as C grows.

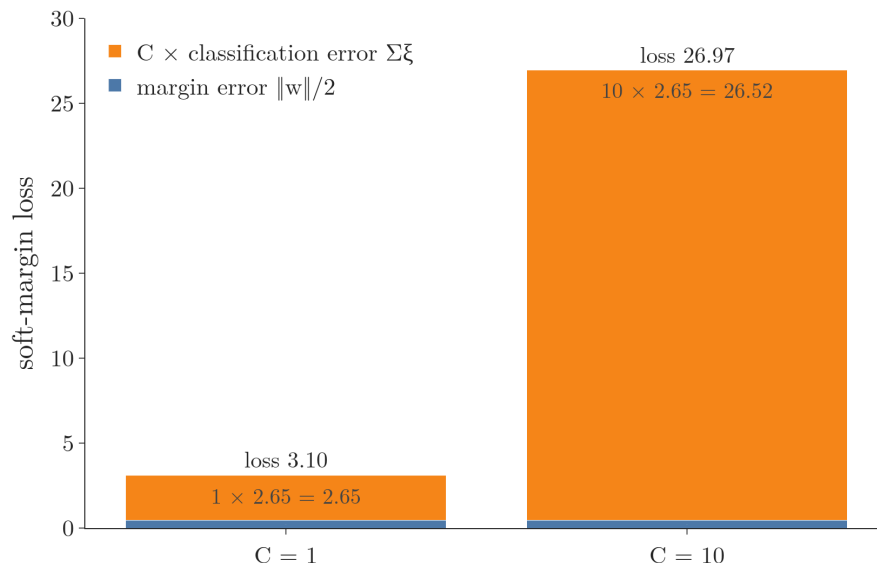


Figure 5: The soft-margin loss of the line of Figure 4 as two stacked parts: margin error 0.45 (blue) and C times the classification error 2.65 (orange). $C = 1$ gives 3.10; $C = 10$ gives 26.97.

6. The hyperparameter C

Key point: Large C : avoid mistakes, accept a narrow margin. Small C : keep the margin wide, accept some mistakes. Tune C with cross-validation.

C (G-336) is a **hyperparameter** (G-910): a tuning knob we set before training. C can be any positive number. An everyday picture: C is the size of the fine for each rule-breaking point. A huge fine makes the line bend over backwards to avoid any mistake; a tiny fine lets it relax and keep a wide road.

- **Large C** (say 1,000 or 10,000): the classification error dominates. The algorithm stops caring about the width of the margin and concentrates on misclassifying no point.
- **Small C** (say 0.1, 0.01 or 0.001): the margin error dominates. The algorithm keeps the margin wide even if some points end up misclassified.

Figure 6 lowers C step by step on the same data and pauses at three values. Watch the margin widen as the fine for mistakes falls, while more points need slack (orange). With $C = 1000$ the margin is only 0.37 wide, but no point is misclassified. With $C = 1$ it is 2.22 wide with two mistakes; with $C = 0.05$ it is 4.65 wide.

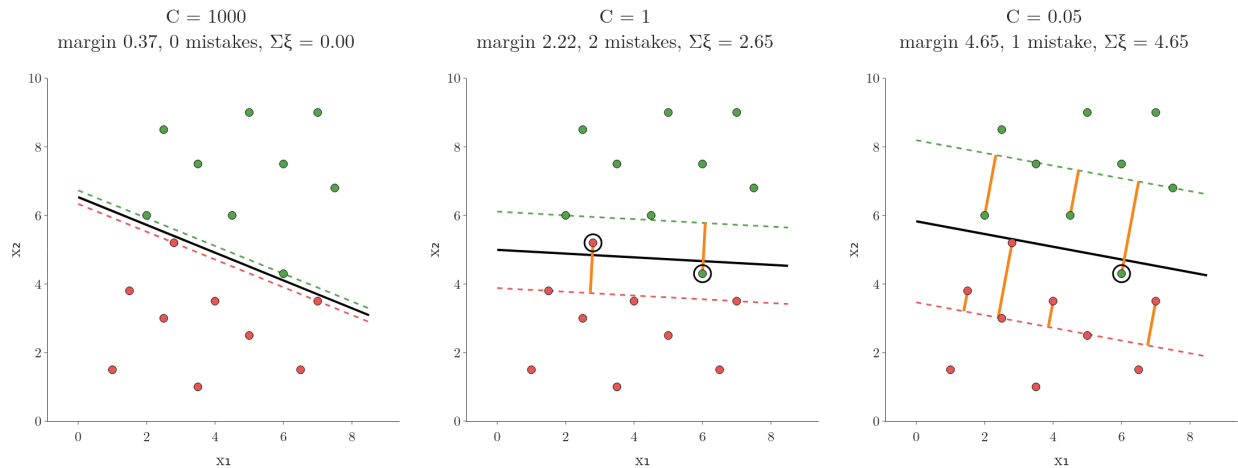


Figure 6: C falling from 1000 to 0.03. Large C : a narrow margin and no mistakes. Small C : a wide margin and some mistakes (ringed); orange sticks are the slacks ξ

Neither extreme is right in general. A model with no training mistakes and a thin margin may be fitting outliers; a model with a very wide margin may ignore real structure. So we pick C for each dataset with cross-validation (ISL §9.2.2), usually through a grid search over values such as 0.001, 0.01, ..., 1,000.

Python: In scikit-learn, C is an argument of `SVC`; its default is 1.

```
from sklearn.svm import SVC
from sklearn.model_selection import GridSearchCV

svm = SVC(kernel="linear", C=1)      # soft margin, C = 1
svm.fit(X, y)

# choose C by 5-fold cross-validation
grid = GridSearchCV(SVC(kernel="linear"),
                    {"C": [0.001, 0.01, 0.1, 1, 10, 100, 1000]}, cv=5)
grid.fit(X, y)
grid.best_params_
```

7. Comparison with logistic regression

Key point: Both are “loss + regularisation”. SVM uses the hinge loss where logistic regression uses the log loss, and in both, C is the inverse of the regularisation strength λ .

The soft-margin loss has the same shape as the loss of regularised logistic regression:

	Error on the data	Regularisation	Strength
Logistic regression	log loss	$\lambda \ w\ ^2$ (L2)	λ on the penalty
Soft-margin SVM	$\sum \xi_i$: the hinge loss	$\ w\ /2$	C on the error

- The margin term $\|w\|/2$ plays the role of the **regularisation** (G-1659) term: it keeps w small, which here means a wide margin.
- The error term $\sum \xi_i$ is called the **hinge loss** (G-898), the SVM’s counterpart of the log loss.
- C multiplies the error instead of the penalty, so it works the other way round: written as “hinge loss + $\frac{\lambda}{2}\|w\|^2$ ”, the SVM has $\lambda = 1/C$ (ESL §12.3.2). A large C means weak regularisation; a small C means strong regularisation.

scikit-learn’s `LogisticRegression` also has a C and no λ (see the [logistic regression hyperparameters Note](#)), following the same convention as SVM: a larger C means less regularisation in both (scikit-learn docs, `LogisticRegression`).

Extra: Written with $z = y_i(w^T x_i + b)$, the hinge loss of one point is $\max(0, 1 - z)$ and the log loss is $\log(1 + e^{-z})$ (the same log loss as before, with target values ± 1). Figure 7 compares them. The hinge loss is exactly 0 for every point beyond its own hyperplane ($z \geq 1$), so those points do not affect the line at all; only points on or inside the margin matter, which is why SVM depends only on its support vectors. The log loss never quite reaches 0, so every point keeps pulling a little.

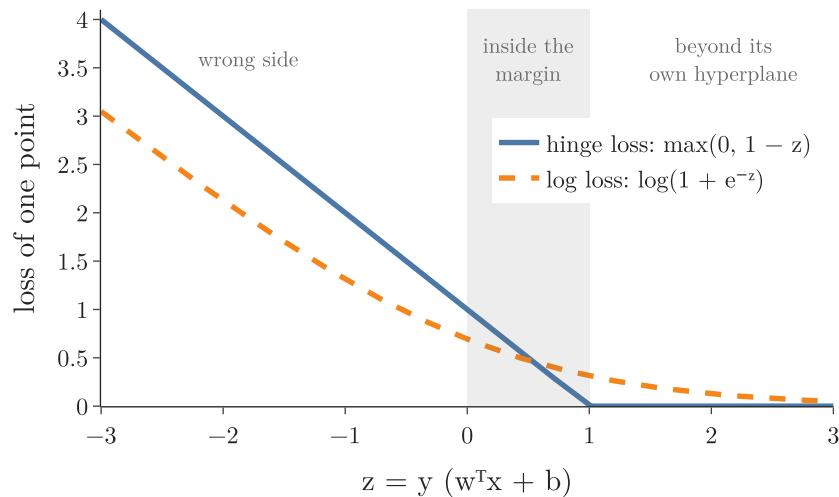


Figure 7: Hinge loss and log loss for one point, against $z = y(w^T x + b)$

8. Why “soft margin”

Key point: The margin is “soft” because points may enter it or cross it, at a price.

The hard-margin SVM forbids any point inside the margin. The soft-margin SVM relaxes that rule: it allows some errors, as long as it pays for them through $C \sum \xi_i$. Because the criterion is softened, this formulation is called the **soft-margin SVM**. The soft-margin SVM is the version scikit-learn’s `SVC` uses.

9. Summary

	Hard-margin SVM	Soft-margin SVM
Objective	$\min \ w\ /2$	$\min \ w\ /2 + C \sum \xi_i$
Points inside the margin or misclassified	not allowed	allowed, at a cost ξ_i
Works on	perfectly separable data	almost separable data with outliers
Knob	none	C

- $\arg \max 2/\|w\|$ equals $\arg \min \|w\|/2$.
- Slack ξ_i : 0 on the correct side of a point's own hyperplane, otherwise how far the point is past it, in margin units ($\xi_i > 1$: wrong side of π).
- Loss = margin error + $C \times$ classification error (hinge loss).
- Large C: narrow margin, few mistakes. Small C: wide margin, more mistakes. Tune C by cross-validation.
- C plays the role of $1/\lambda$, as in `LogisticRegression`.

10. Sources

Built from

- CampusX, “Mathematics of Support Vector Machine | Soft Margin SVM”, YouTube, <https://www.youtube.com/watch?v=utqrvIFAE1k>
- StatQuest with Josh Starmer, “Support Vector Machines Part 1 (of 3): Main Ideas!!!”, YouTube, <https://www.youtube.com/watch?v=efR1C6CvbmE> (the one-dimensional outlier, the soft margin and the bias-variance reason for it; Figure 2 recreates it on iris petal lengths)

Other references

- **ISL**: James, G., Witten, D., Hastie, T. and Tibshirani, R. *An Introduction to Statistical Learning*, 2nd edition. Springer, 2021. Sections 9.1.5 (p. 373) and 9.2.2 (p. 377).
- **ESL**: Hastie, T., Tibshirani, R. and Friedman, J. *The Elements of Statistical Learning*, 2nd ed. Springer, 2009. Section 12.3.2, p. 426.
- **scikit-learn docs**: user guide Section 1.4.7, “Mathematical formulation” (SVC); `sklearn.linear_model.LogisticRegression` (parameter C), scikit-learn 1.9.

11. Key terms

Term	Meaning
Soft-margin SVM	The SVM that allows points inside the margin or on the wrong side, at a cost controlled by C

Term	Meaning
Slack (ξ)	How far a training point lies on the wrong side of its own hyperplane; 0 if it is on the correct side
Margin error	The term $\ w\ /2$ of the SVM loss; small when the margin is wide
Classification error (SVM)	The term $\sum \xi_i$ of the SVM loss: the total slack of all points
Hinge loss	$\max(0, 1 - y(w^T x + b))$ per point: the error term of the soft-margin SVM
C (SVM)	The weight on the classification error; a large C means few mistakes and a narrow margin, a small C a wide margin